Mukesh Kumar Sharma

Senior iOS Software Engineer | Swift, SwiftUI, Swift Concurrency, Mobile Architecture, On-Device AI
London, ON, Canada | Open to relocation | +1 519-859-0693 | mukesh.ct.pce@gmail.com
linkedin.com/in/imuks | github.com/imuks | imuks.github.io

SUMMARY

Senior iOS engineer with 15+ years building and shipping apps across banking, retail, enterprise payments, and smart-city platforms, including apps used by millions of customers at TD, Kroger, and Gap; 10+ years in Swift, 5+ in SwiftUI. Designs reusable frameworks other teams adopt (an SPM framework used by ~20 apps), owns CI/CD and release quality, mentors engineers, and works closely with product, design, and backend. Most recently built the real-time LLM streaming layer of Vanguard's generative-AI investing assistant, and independently shipped an on-device AI healthcare product using Apple Foundation Models fine-tuned with LoRA adapters.

TECHNICAL SKILLS

Programming Languages: Swift 6 and 5.x, Objective-C, TypeScript, JavaScript, Ruby (Fastlane)
iOS & UI: SwiftUI, UIKit (programmatic), Combine, Observation (@Observable), Core Data, Core Animation, Core Graphics, AVFoundation, MapKit, WidgetKit, App Intents, TipKit, Accessibility (VoiceOver, Dynamic Type)
Concurrency: async/await, actors, @globalActor, Sendable, structured concurrency, strict concurrency checking, GCD
Architecture: MVVM with Coordinator, Clean/VIP, protocol-oriented design, dependency injection, repository pattern, SOLID, SPM modularization (25 internal packages)
On-Device & Applied AI: Apple Foundation Models, Core ML, LoRA adapters, LLM response streaming (SSE, newline-delimited JSON), Apple Speech, OpenAI API, prompt engineering
Networking & Security: URLSession, Alamofire, REST, GraphQL (Apollo iOS), WebSockets, Codable, OAuth 2.0, Keychain, Face ID, Touch ID, SSL pinning, APNs, Firebase
Testing & Performance: XCTest, Swift Testing, XCUITest, TDD, Instruments (Time Profiler, Allocations, Leaks)
CI/CD & Release: Fastlane, GitHub Actions, Xcode Cloud, TestFlight, feature flags, A/B testing, Crashlytics, Sentry
Cross-Platform & Backend: React Native with native Swift modules, Node.js/TypeScript REST APIs on Google App Engine

PROFESSIONAL EXPERIENCE

Senior iOS Engineer
Mar 2026 – Aug 2026

Vanguard (contract via Publicis Sapient) | Toronto, ON (hybrid)

- Delivered the generative-AI investing assistant in Vanguard's flagship iOS app across 8 Swift/SwiftUI modules, including a reusable SwiftUI chat component package, through the phase 1 production release (staged rollout).
- Implemented real-time LLM streaming so clients see responses as they're generated (Alamofire DataStreamRequest, a custom newline-delimited JSON tokenizer, Combine), and eliminated data races under load with an actor-isolated tokenizer, a custom @globalActor, @MainActor UI boundaries, and full Sendable conformance.
- Co-designed the streaming contract with backend and AI-platform teams and aligned iOS, backend, and design on its schema: 8 response types (markdown, tables, lists, charts) with custom Decodable parsing and graceful degradation.
- Built a reusable coach-mark framework modeled on TipKit and back-deployed to iOS 17, plus an animated onboarding flow to guide first-time users; adopted by other engineers and validated through iOS 26.
- Modularized the feature into SPM packages with a protocol and paired mock per ViewModel, tested the tokenizer, API decoding, and ViewModels with XCTest and Swift Testing, and built custom Claude commands that automate first-pass code review against team Swift/SwiftUI guidelines.

Lead iOS Engineer
Dec 2024 – Mar 2026

Tech Mahindra | Enterprise payments platform (iOS) and TruGreen consumer app (React Native)

- Led 0-to-1 delivery of the client's SwiftUI payments app, the checkout hub for several business lines; owned MVVM architecture, checkout, transaction history, and merchant onboarding, and ran design reviews with product, design, and QA.
- Designed an offline transaction queue (Core Data, BGTaskScheduler, 72-hour retention) so field sales could take payments without a signal and sync on reconnect, reducing offline transaction failures to effectively zero; integrated Bluetooth card readers and MFi accessories with background modes and a retry layer.
- Built a reusable Swift Package (payments, receipts, deep linking) adopted by 3 internal apps, ending per-app rewrites.
- Owned CI/CD and cut release cycle time 20–50% by automating TestFlight and App Store releases (Fastlane, GitHub Actions, Xcode Cloud); reached 80% unit-test coverage on business logic and used Instruments to fix retain cycles and improve cold start and memory pressure.
- In parallel, shipped TruGreen features in React Native (TypeScript) to iOS and Android from one codebase, writing native Swift modules for biometrics, deep links, and secure storage that JavaScript couldn't reach.

Founder & Lead Engineer
Jan 2024 – Present

Dr Godly, on-device AI healthcare product (iOS, Android, Node.js) | Limited production pilot with early customers

- Fine-tuned Apple Foundation Models with LoRA adapters for clinical vocabulary and note summarization, with default safety guardrails; the model runs entirely on device, so patient audio and transcripts never reach a server for AI processing.

- Added live two-party transcription to video consultations: real-time speech-to-text for doctor and patient during Twilio Video calls, plus text-to-speech playback, using Apple Speech and the LoRA-tuned on-device model.
- Built and shipped the full stack solo across 6+ repositories: iOS client, Android companion, shared Swift Package, and a Node.js/TypeScript backend on Google App Engine (auth, sessions, LLM orchestration).

Senior / Lead iOS Engineer Jun 2023 – Jan 2024
NEOM (via Tech Mahindra) | Smart City, Saudi Arabia

- Architected and shipped the NEOM Smart City app's first App Store release for residents, visitors, and service providers.
- Designed the core SPM framework (networking, OAuth, SSL pinning, shared UI) adopted by ~20 internal NEOM apps, including its interface contracts and versioning strategy; integrated Apollo iOS for type-safe GraphQL.
- Integrated the OpenAI iOS client and a LangChain-based prompt SDK to power an in-app assistant for utilities, transport, and support; added Twilio audio/video for resident-to-agent support and Apple Speech for voice interaction.
- Mentored 3 engineers and led a small iOS team through the full lifecycle to a production launch, running design reviews with product, backend, and security and owning architectural decisions.

Senior / Lead iOS & Release Engineer Jan 2023 – Jun 2023
Kroger (via Tech Mahindra)

- Owned end-to-end iOS delivery of the Kroger Gamification module (v1.0 and v2.0), released to the full customer base across Kroger, Ralphs, Fred Meyer, and Harris Teeter.
- Authored the build, sign, and deploy pipelines adopted by the entire iOS org across 4 Kroger banner apps (Fastlane, GitHub Actions, shell, Ruby, Node.js).
- Instrumented rollouts with Firebase Remote Config flags and A/B experiments, monitored via Grafana and Crashlytics.

Senior iOS Engineer Jun 2022 – Dec 2022
Gap Inc. (via Avenue Code)

- Delivered features for 4 storefronts (Gap, Athleta, Banana Republic, Old Navy) in Swift, SwiftUI, and Objective-C.
- Led the Notification System workstream from scratch to production launch with product and backend partners.

Senior iOS Engineer 2021 – 2022
TD Bank (via Clearbridge Mobile / Amdocs)

- Led 3 core banking modules in Swift and Objective-C, shipping features used by millions of TD banking customers.
- Integrated Face ID and Touch ID sign-in (LocalAuthentication, Keychain) and hardened the app against jailbreak and tampering; contributed to TD's iOS Banking Foundation Framework, distributed via CocoaPods.

Lead iOS Engineer 2017 – 2021
Tata Consultancy Services | Smart-utility and Green Button apps

- Owned the iOS codebase for a smart-utility app (real-time energy gauges, device control, push) on an AWS backend.
- Built cross-platform Green Button apps for 3 utilities (Toronto Hydro, London Hydro, Whitby Hydro) that show customers their own energy-use data.

iOS Engineer 2011 – 2017
Pfizer (via Tata Consultancy Services), GlobalLogic, smartData, Regard Solutions

- Shipped 6 internal Pfizer apps (Objective-C, Swift) with custom Core Graphics charts, Janrain SSO, and offline sync.
- Shipped 8+ App Store apps across healthcare, retail, and consumer, and contributed accessibility R&D to the Qualcomm Healthy Circle proof of concept.

OPEN SOURCE

Open-source iOS and AI projects | github.com/imuks 2023 – Present

- chatGPT_iOS: Swift OpenAI client with streaming and SwiftUI (January 2023)
- StableDiffusion_iOS: on-device image generation
- swift_DSA: data structures and algorithms reference in Swift

EDUCATION

B.E., Computer Technology | Priyadarshini College of Engineering, R.T.M. Nagpur University 2011